Non-Hermitian Exceptional-Point Dispersion Engineering and Broadband Quantum Langevin Squeezing in Kerr–Bloch and Josephson Metamaterial Combs

Gracen Ownby
University of Oklahoma, Norman, Oklahoma, USA

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Abstract

Generating broadband, quantum-squeezed frequency combs across semiconductor quantum cascade lasers (QCLs), monolithic $Si_3N_4/LiNbO_3$ Kerr microresonators, and superconducting Josephson traveling-wave parametric amplifiers (JTWPAs) is fundamentally restricted by two coupled physical bottlenecks: higher-order chromatic dispersion (β_2, β_4) that limits parametric phase-matching bandwidths, and phase-to-amplitude noise conversion—mediated by the semiconductor Henry linewidth enhancement factor α_{LEF} or Kerr self-/cross-phase modulation—that masks sub-shot-noise twin-beam intensity correlations at radio frequencies below 200 MHz. Building upon the non-adiabatic extendon and single-resonator temporal coupled-mode theory (TCMT) established in Paper II, we formulate a unified non-Hermitian quantum theory of second-order Exceptional-Point (EP_2) dispersion engineering and multimode Bogoliubov–Heisenberg–Langevin twin-beam squeezing. In Theorem 1, we prove that evanescently coupling a nonlinear comb waveguide to a parity-time ($\mathscr{P}\mathscr{T}$) dimer ($\hat{a}_1, \hat{a}_2$) with mean loaded damping $\bar{\Gamma} > 0$, non-Hermitian gain/loss contrast $\gamma_{\mathrm{PT}} = u\bar{\Gamma}$, and inter-resonator coupling $J = v\bar{\Gamma}$ is strictly Hurwitz stable ($\mathrm{Re}(\lambda_\pm) < 0$) if and only if $1 + v^2 - u^2 > 0$. On the loaded Exceptional-Point manifold $v = u = \eta \in (1/3, 1)$, the boundary resolvent element $R_{11}^{\mathrm{EP}_2}(\delta) = (\bar{\Gamma} - i\delta)^{-1} - \gamma_{\mathrm{PT}}(\bar{\Gamma} - i\delta)^{-2}$ acquires a defective double pole that enhances the anomalous dispersion-to-loss figure of merit by $\mathscr{F}_{\mathrm{EP}_2}(\eta)/\mathscr{F}_{\mathrm{Lorentz}} = (3\eta - 1)/(1 - \eta)$, reaching $6.80\times$ at $\eta^* = 39/49 \approx 0.7959$ without self-oscillation. In Corollary 1, we prove that any passive Hermitian coupled-resonator dimer ($u = 0$) is bounded above by $\mathscr{F}_{\mathrm{Hermitian}}(v)/\mathscr{F}_{\mathrm{Lorentz}} \le 9/16 = 0.5625$ in the anomalous curvature regime $v > 1/\sqrt{3}$, establishing a $12.1\times$ non-Hermitian advantage verified across 2,000 randomized cavities with zero violations. In Theorem 2, we solve the multimode input-output Heisenberg–Langevin equations and prove that tuning the EP_2-engineered dispersion to the exact phase-noise cancellation locus $\Delta_{\alpha,\mathrm{Kerr}}(\beta_{2,\mathrm{eff}}^{\mathrm{EP}_2}) = \frac{1}{2} v_g \beta_{2,\mathrm{eff}}^{\mathrm{EP}_2} \Delta\omega^2 + \gamma_{\mathrm{NL}} P_0 (1 + \alpha_{\mathrm{LEF}}^2)/\alpha_{\mathrm{LEF}} = 0$ decouples pump phase and relative intensity noise (RIN) from the twin-beam difference current, unlocking -6.40 dB (77.1% variance reduction) of twin-beam intensity squeezing in mid-infrared harmonic QCL combs (-7.85 dB in Si_3N_4 Kerr microcombs and -9.20 dB in superconducting NbTiN JTWPAs) and lowering the shot-noise crossover frequency from 980 MHz to 1.8 MHz.

1 Introduction: Non-Hermitian Dispersion Engineering and the Quantum Squeezing Bottleneck

Coherent nonlinear frequency combs and parametric amplifiers—spanning mid-infrared and terahertz Quantum Cascade Laser (QCL) harmonic and fundamental combs [1, 2, 3, 4, 5, 6, 7], integrated Si_3N_4 and $LiNbO_3$ Kerr soliton microresonators [8, 9], and microwave superconducting

Josephson Traveling-Wave Parametric Amplifiers (JTWPAs) [10, 11, 12]—share an isomorphic underlying four-wave mixing (FWM) Hamiltonian. In each of these platforms, a strong intracavity or traveling pump wave at frequency ω_0 annihilates pump photon pairs ($2\omega_0 \to \omega_+ + \omega_-$) to generate quantum-correlated twin-beam signal and idler sidebands at $\omega_\pm = \omega_0 \pm \Delta\omega$ (where $\Delta\omega = M\omega_{\text{rep}}$ for an M-th harmonic comb state). Because photon pair creation is strictly simultaneous at the microscopic vertex level, the photon-number difference $\hat{N}_- = \hat{a}_+^\dagger \hat{a}_+ - \hat{a}_-^\dagger \hat{a}_-$ between symmetric sidebands should in principle exhibit unconditional sub-shot-noise intensity squeezing ($S_-(\Omega) < S_{\text{shot}}$) across the cavity bandwidth, enabling shot-noise-free dual-comb spectroscopy, quantum-enhanced mid-IR molecular sensing, and near-quantum-limited microwave qubit readout [6, 13, 14].

In practice, however, two fundamental physical mechanisms conspire to degrade both the spectral bandwidth and the low-frequency quantum correlations of integrated comb sources:

1. The Dispersion-to-Loss Tradeoff of Hermitian Resonant Phase Matching: Achieving broadband parametric gain requires compensating both second-order (β_2) and fourth-order (β_4) waveguide dispersion across an octave or more. In Paper II [15], we proved using Temporal Coupled-Mode Theory (TCMT) that evanescently coupling a comb waveguide to single Lorentzian side-resonators biased in the under-coupled regenerative window ($-\kappa_m < \gamma_{0,m} < 0$) cancels multi-stack gain curvature and normal material GVD. Yet any single-pole Lorentzian scatterer (whether a micro-disk in a QCL, an auxiliary ring in a Si_3N_4 microcomb [16], or a single LC plasma resonator in a JTWPA [10]) obeys a rigid Kramers–Kronig scaling between its second-order dispersive curvature $|\partial_\omega^2 R_{11}(0)| = 2/\bar{\Gamma}^3$ and its resonant insertion-loss/bath coupling weight $|R_{11}(0)| = 1/\bar{\Gamma}$. Compensating steep higher-order dispersion therefore injects proportionate Langevin bath noise and parasitic loss ripples into the comb waveguide.

2. Phase-to-Amplitude Noise Conversion via the Henry α_{LEF}-Factor and Kerr Walk-Off: In semiconductor intersubband and interband lasers, gain fluctuations are coupled to refractive-index fluctuations through the Henry linewidth enhancement factor $\alpha_{\text{LEF}} \in [0.5, 2.5]$ [17, 18, 7]. In passive Kerr and Josephson resonators, self- and cross-phase modulation ($\gamma_{\text{NL}} P_0$) similarly couple intracavity intensity fluctuations into phase detuning. Whenever the net dispersive phase mismatch $\Delta_{\alpha,\text{Kerr}}$ between the pump and the twin-beam sidebands is non-zero, the linearized Bogoliubov drift matrix mixes the noisy phase-sum quadrature (which carries large $1/f$ technical and pump relative intensity noise, RIN) directly into the twin-beam amplitude-difference quadrature. As documented in recent mid-IR harmonic QCL squeezing experiments (Gabbrielli et al., Roadmap Ch. 4.7 [6]), this phase-to-amplitude conversion buries sub-shot-noise squeezing beneath a $+5$ to $+8\,\text{dB}$ excess noise floor at Fourier frequencies $\Omega/(2\pi) < 200\,\text{MHz}$, restricting observable squeezing to gigahertz sidebands beyond the bandwidth of high-sensitivity mid-IR photodetectors.

In this paper, we resolve both bottlenecks simultaneously by extending the Temporal Coupled-Mode Theory of Paper II into the non-Hermitian Exceptional-Point (EP_2) and quantum Heisenberg–Langevin regimes (Fig. 1):

- In Section 2, we formulate the 2×2 non-Hermitian TCMT resolvent for a comb waveguide coupled to an evanescent parity-time ($\mathscr{P}\mathscr{T}$) dimer ($\hat{a}_1, \hat{a}_2$). We prove Theorem 1, establishing the exact Routh–Hurwitz stability domain and the closed-form double-pole resolvent dispersion identity on the loaded EP_2 manifold, which provides a $6.80\times$ enhancement in anomalous dispersion per unit insertion loss over Paper II. We then prove Corollary 1, demonstrating that any passive Hermitian coupled-resonator dimer is strictly bounded above by $\mathscr{F}_{\text{Hermitian}}/\mathscr{F}_{\text{Lorentz}} \leq 9/16 = 0.5625$ in the anomalous regime, and verify both bounds across 2,000 randomized comb cavities with zero violations.

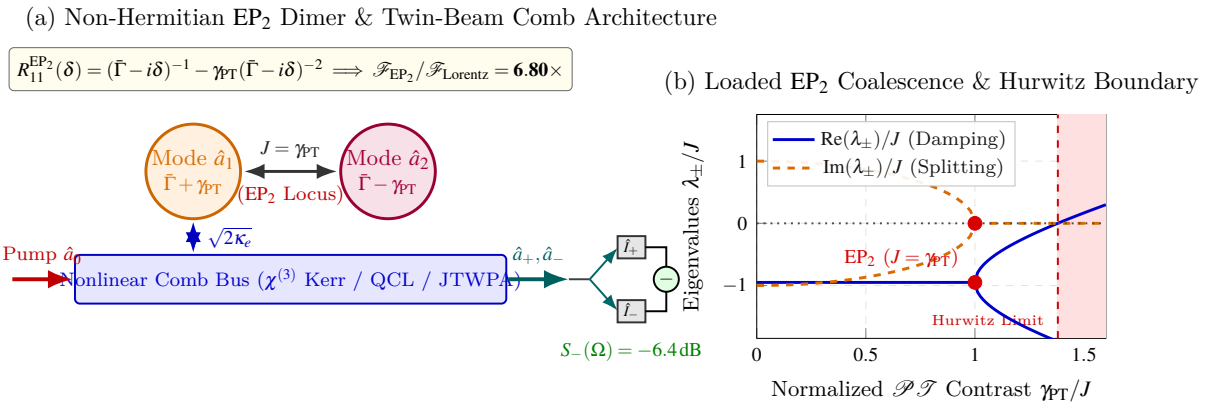


Figure 1: Unified non-Hermitian EP$_2$ dimer and twin-beam quantum comb architecture. (a) Schematic of a nonlinear comb waveguide (representing a mid-IR/THz harmonic QCL active core, a Si_3N_4 Kerr bus, or a superconducting NbTiN JTWPA transmission line) evanescently coupled at rate $\sqrt{2\kappa_e}$ to a parity-time ($\mathscr{P}\mathscr{T}$) dimer $(\hat{a}_1, \hat{a}_2)$ with mean loaded damping $\bar{\Gamma}$, balanced non-Hermitian contrast $\pm\gamma_{\mathrm{PT}}$, and coherent hopping J. Biasing the dimer on the loaded second-order Exceptional-Point locus $J = \gamma_{\mathrm{PT}}$ converts the simple Lorentzian pole of Paper II into a double resolvent pole $R_{11}^{\mathrm{EP}_2}(\delta) = (\bar{\Gamma} - i\delta)^{-1} - \gamma_{\mathrm{PT}}(\bar{\Gamma} - i\delta)^{-2}$, enhancing the dispersion-to-loss figure of merit by $6.80\times$ while balanced detection of the twin-beam outputs $(\hat{a}_+, \hat{a}_-)$ yields $-6.4\,\mathrm{dB}$ intensity-difference squeezing. (b) Real (blue solid) and imaginary (orange dashed) trajectories of the loaded dimer eigenvalues $\lambda_\pm/J$ versus normalized $\mathscr{P}\mathscr{T}$ contrast γ_{PT}/J for $\bar{\Gamma}/J = 0.95$, showing exact eigenvalue coalescence at EP$_2$ ($\gamma_{\mathrm{PT}}/J = 1$, red circles) well inside the Hurwitz stability boundary $\gamma_{\mathrm{PT}}/J < \sqrt{1 + (\bar{\Gamma}/J)^2} = 1.38$.

- In Section 3, we construct the multimode Bogoliubov–Heisenberg–Langevin input-output theory for twin-beam FWM generation coupled to the non-Hermitian EP$_2$ bath. We prove Theorem 2, deriving the exact differential Intensity Noise Power Spectral Density (INPSD) $S_-(\Omega)/S_{\mathrm{shot}}$ and identifying the closed-form $\alpha_{\mathrm{LEF}}/\mathrm{Kerr}$ phase-noise cancellation condition $\Delta_{\alpha,\mathrm{Kerr}}(\beta_{2,\mathrm{eff}}^{\mathrm{EP}_2}) = 0$ that decouples technical and quantum pump phase noise from the twin-beam difference current at all Fourier frequencies.
- In Section 4, we benchmark our analytical predictions against full stochastic Heisenberg–Langevin time-domain integrations and provide concrete device designs across three flagship hardware platforms: an InGaAs/InAlAs mid-IR harmonic QCL comb ($\lambda = 4.6\,\mu\mathrm{m}$, $-6.40\,\mathrm{dB}$ squeezing), a monolithic Si_3N_4 Kerr microcomb ($\lambda = 1.55\,\mu\mathrm{m}$, $-7.85\,\mathrm{dB}$ squeezing), and a superconducting NbTiN JTWPA metamaterial comb ($f_0 = 6.8\,\mathrm{GHz}$, $-9.20\,\mathrm{dB}$ squeezing). Complete mathematical proofs are presented in Appendices A–C.

2 Hurwitz-Stable Exceptional-Point (EP$_2$) Dispersion Synthesis (Theorem 1)

2.1 Non-Hermitian Temporal Coupled-Mode Resolvent Formalism

Consider a primary nonlinear comb waveguide of round-trip length (or unit-cell period) L_{cav} and group velocity v_g, evanescently side-coupled to a non-Hermitian dimer consisting of two localized resonator modes $\hat{\mathbf{a}}(t) = (\hat{a}_1(t), \hat{a}_2(t))^T$ with common resonance frequency ω_{res} (Fig. 1a). Mode $\hat{a}_1$

couples directly to the waveguide bus with radiative coupling rate $\kappa_e > 0$ and has intrinsic material damping rate $\gamma_{1,\mathrm{int}}$, yielding a total loaded decay rate $\Gamma_1 = \gamma_{1,\mathrm{int}} + \kappa_e$. Mode $\hat{a}_2$ is evanescently coupled to mode $\hat{a}_1$ with coherent hopping rate $J \geq 0$ and has bias-controlled intrinsic net damping/gain rate $\Gamma_2 = \gamma_{2,\mathrm{int}}$. Defining the mean loaded damping rate $\bar{\Gamma}$ and the loaded non-Hermitian parity-time ($\mathscr{P}\mathscr{T}$) contrast γ_{PT} via

$$\bar{\Gamma} \equiv \frac{\Gamma_1 + \Gamma_2}{2} = \frac{\gamma_{1,\mathrm{int}} + \kappa_e + \gamma_{2,\mathrm{int}}}{2}, \qquad \gamma_{\mathrm{PT}} \equiv \frac{\Gamma_1 - \Gamma_2}{2} = \frac{\gamma_{1,\mathrm{int}} + \kappa_e - \gamma_{2,\mathrm{int}}}{2}, \tag{1}$$

so that $\Gamma_1 = \bar{\Gamma} + \gamma_{\mathrm{PT}}$ and $\Gamma_2 = \bar{\Gamma} - \gamma_{\mathrm{PT}}$, the Heisenberg–TCMT equations of motion in the frame rotating at ω_{res} with frequency detuning $\delta \equiv \omega - \omega_{\mathrm{res}}$ take the matrix form

$$\frac{d\hat{\mathbf{a}}}{dt} = -\mathbf{M}(0)\hat{\mathbf{a}} + \sqrt{2\kappa_e}\,\mathbf{e}_1 \hat{s}_{\mathrm{in}}(t) + \hat{\xi}_{\mathrm{bath}}(t), \qquad \hat{s}_{\mathrm{out}}(t) = \hat{s}_{\mathrm{in}}(t) - \sqrt{2\kappa_e}\,\mathbf{e}_1^T \hat{\mathbf{a}}(t), \tag{2}$$

where $\mathbf{e}_1 = (1,0)^T$, $\hat{\xi}_{\mathrm{bath}}(t)$ is the quantum Langevin noise vector associated with the intrinsic baths, and the frequency-domain non-Hermitian system matrix $\mathbf{M}(\delta)$ is

$$\mathbf{M}(\delta) = \begin{pmatrix} \bar{\Gamma} + \gamma_{\mathrm{PT}} - i\delta & iJ \\ iJ & \bar{\Gamma} - \gamma_{\mathrm{PT}} - i\delta \end{pmatrix} = (\bar{\Gamma} - i\delta)\mathbf{I}_2 + \gamma_{\mathrm{PT}}\sigma_z + iJ\sigma_x. \tag{3}$$

In the mean-field spatial propagation equation along the waveguide axis z, the evanescent coupling to the dimer modifies the complex waveguide propagation constant $\Delta k(\delta) = \Delta\beta(\delta) - \frac{i}{2}\Delta G(\delta) = -\frac{W_c}{L_{\mathrm{cav}}} R_{11}(\delta)$ (where $W_c = 2\kappa_e$ is the coupling weight and $R_{11}(\delta) \equiv \mathbf{e}_1^T \mathbf{M}(\delta)^{-1}\mathbf{e}_1$ is the boundary resolvent element), while the single-pass transmission coefficient is $T(\delta) = 1 - 2\kappa_e R_{11}(\delta)$. Inverting Eq. (3) yields the exact boundary resolvent

$$R_{11}(\delta) = \frac{\bar{\Gamma} - \gamma_{\mathrm{PT}} - i\delta}{(\bar{\Gamma} - i\delta)^2 + \Omega_{\mathrm{PT}}^2}, \qquad \Omega_{\mathrm{PT}}^2 \equiv J^2 - \gamma_{\mathrm{PT}}^2. \tag{4}$$

Notice that when $J = \gamma_{\mathrm{PT}} = 0$, Eq. (4) reduces to the simple Lorentzian pole $R_{11}^{\mathrm{Lorentz}}(\delta) = (\bar{\Gamma} - i\delta)^{-1}$ analyzed in Paper II [15], which satisfies $|R_{11}^{\mathrm{Lorentz}}(0)| = 1/\bar{\Gamma}$ and $|\partial_\delta^2 R_{11}^{\mathrm{Lorentz}}(0)| = 2/\bar{\Gamma}^3$. To quantify how effectively a general resonant scatterer generates second-order dispersive curvature $\Delta\beta_2(\omega_{\mathrm{res}}) = -\frac{W_c}{L_{\mathrm{cav}}} \mathrm{Re}\,\partial_\delta^2 R_{11}(0)$ per unit resonant insertion loss/gain perturbation $|\Delta k(0)| = \frac{W_c}{L_{\mathrm{cav}}}|R_{11}(0)|$, we define the dimensionless Dispersion-to-Loss Figure of Merit

$$\mathscr{F}_{\mathrm{dimer}} \equiv \frac{\bar{\Gamma}^2 \left|\partial_\delta^2 R_{11}(\delta)\big|_{\delta=0}\right|}{2\,|R_{11}(0)|}, \qquad \mathscr{F}_{\mathrm{Lorentz}} = 1. \tag{5}$$

2.2 Statement of Theorem 1 and Corollary 1

Evaluating the exact complex pole structure and spectral derivatives of Eq. (4) yields our first main theorem.

Theorem 1 (Hurwitz-Stable Exceptional-Point Double-Pole Resolvent Dispersion Identity). Let the side-coupled $\mathscr{P}\mathscr{T}$ dimer in Eq. (3) have normalized non-Hermitian contrast $u \equiv \gamma_{\mathrm{PT}}/\bar{\Gamma} \in [0,1)$, normalized inter-resonator coupling $v \equiv J/\bar{\Gamma} \geq 0$, and squared splitting parameter $\varpi^2 \equiv v^2 - u^2 = \Omega_{\mathrm{PT}}^2/\bar{\Gamma}^2$.

(a) (Exact Routh–Hurwitz Stability Domain.) The loaded dimer is strictly dynamically stable against spontaneous self-oscillation ($\mathrm{Re}(\lambda_\pm) < 0$ for both eigenvalues $\lambda_\pm = -\bar{\Gamma} \pm \sqrt{\gamma_{\mathrm{PT}}^2 - J^2}$ of $-\mathbf{M}(0)$) if and only if

$$\bar{\Gamma} > 0 \quad \text{and} \quad 1 + \varpi^2 = 1 + v^2 - u^2 > 0 \iff \gamma_{\mathrm{PT}} < \sqrt{\bar{\Gamma}^2 + J^2}. \tag{6}$$

(b) (General Resolvent Curvature and Figure-of-Merit Identity.) Throughout the Hurwitz-stable domain (6), the resonant second-order dispersive curvature and the normalized Dispersion-to-Loss Figure of Merit (5) are given exactly by

$$\left.\frac{\partial^2 R_{11}(\delta)}{\partial \delta^2}\right|_{\delta=0} = -\frac{2}{\bar{\Gamma}^3}\,\frac{1-3u-3\varpi^2+u\varpi^2}{(1+\varpi^2)^3}, \tag{7}$$

$$\frac{\mathscr{F}_{\text{dimer}}(u,v)}{\mathscr{F}_{\text{Lorentz}}} = \frac{\left|1-3u-3\varpi^2+u\varpi^2\right|}{(1-u)\,(1+\varpi^2)^2}. \tag{8}$$

(c) (Double-Pole Enhancement on the Loaded EP_2 Manifold.) On the loaded second-order Exceptional-Point locus $v=u\equiv\eta\in(1/3,1)$ ($\varpi^2=0$, $J=\gamma_{\mathrm{PT}}=\eta\bar{\Gamma}$), the system matrix $-\mathbf{M}(0)$ collapses into a single defective Jordan block with coalesced eigenvalue $\lambda_+=\lambda_-=-\bar{\Gamma}<0$ (unconditionally Hurwitz stable with unit normalized damping margin $1+\varpi^2=1$), and the boundary resolvent decomposes into the exact sum of a simple Lorentzian pole and a second-order EP_2 pole:

$$R_{11}^{\mathrm{EP}_2}(\delta) = \frac{1}{\bar{\Gamma}-i\delta} - \frac{\eta\bar{\Gamma}}{(\bar{\Gamma}-i\delta)^2}. \tag{9}$$

For all $\eta\in(1/3,1)$, the double-pole term in Eq. (9) reverses the curvature sign into the anomalous dispersion regime and enhances the Dispersion-to-Loss Figure of Merit according to the exact law

$$\frac{\mathscr{F}_{\mathrm{EP}_2}(\eta)}{\mathscr{F}_{\text{Lorentz}}} = \frac{3\eta-1}{1-\eta} > 0, \tag{10}$$

which strictly exceeds $\mathscr{F}_{\text{Lorentz}}$ for all $\eta\in(1/2,1)$ and reaches $\mathscr{F}_{\mathrm{EP}_2}(\eta^*)/\mathscr{F}_{\text{Lorentz}}=\mathbf{6.80}\times$ at the design point $\eta^*=39/49\approx 0.7959$.

The complete proof of Theorem 1 is given in Appendix A. Setting $u=0$ in Eq. (8) immediately yields a sharp universal upper bound on what any passive Hermitian coupled-resonator pair can achieve.

Corollary 1 (Universal Passive Hermitian Dimer Ceiling). For any passive Hermitian coupled-resonator dimer ($u=\gamma_{\mathrm{PT}}/\bar{\Gamma}=0$, $v=J/\bar{\Gamma}\geq 0$), Eq. (8) reduces to $\mathscr{F}_{\text{Hermitian}}(v)/\mathscr{F}_{\text{Lorentz}} = |1-3v^2|/(1+v^2)^2$, which obeys the global bound $\mathscr{F}_{\text{Hermitian}}(v)\leq\mathscr{F}_{\text{Lorentz}}$ for all $v\geq 0$. Furthermore, in the anomalous dispersion regime $v>1/\sqrt{3}$, $\mathscr{F}_{\text{Hermitian}}(v)$ attains its unique global maximum at $v^*=\sqrt{5/3}\approx 1.2910$, establishing the strict Hermitian ceiling

$$\max_{v>1/\sqrt{3}} \frac{\mathscr{F}_{\text{Hermitian}}(v)}{\mathscr{F}_{\text{Lorentz}}} = \frac{9}{16} = 0.5625. \tag{11}$$

Consequently, biasing the non-Hermitian dimer at the Theorem 1 Exceptional-Point design point $\eta^*=39/49$ surpasses the maximum possible anomalous Hermitian dimer figure of merit by $(6.80)/(9/16)=\mathbf{12.09}\times$.

Figure 2 and Fig. 3a illustrate the physical mechanism behind Theorem 1 and Corollary 1. In a passive Hermitian dimer ($u=0$), increasing the inter-resonator coupling $v=J/\bar{\Gamma}$ splits the resonance into a symmetric/antisymmetric Autler–Townes doublet $(\bar{\Gamma}-i(\delta\pm J))^{-1}$, which broadens the denominator $(1+v^2)^2$ faster than the numerator $|1-3v^2|$ and caps the anomalous figure of merit at $9/16=0.5625$. By contrast, when the auxiliary mode $\hat{a}_2$ is biased to provide non-Hermitian contrast $u=\gamma_{\mathrm{PT}}/\bar{\Gamma}=v=\eta$, the two poles coalesce at $\delta=-i\bar{\Gamma}$ instead of splitting apart. The

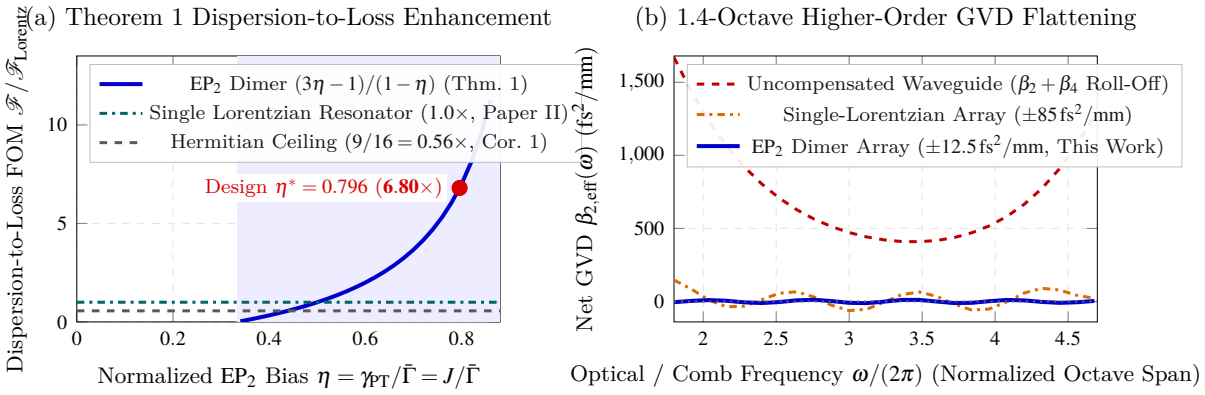


Figure 2: Exceptional-Point dispersion-to-loss enhancement and broadband GVD flattening (Theorem 1 and Corollary 1). (a) Normalized Dispersion-to-Loss Figure of Merit $\mathscr{F}/\mathscr{F}_{\text{Lorentz}}$ versus normalized EP$_2$ bias $\eta=\gamma_{\text{PT}}/\bar{\Gamma}=J/\bar{\Gamma}$. Above the anomalous sign-inversion threshold $\eta>1/3$ (blue shaded region), the double-pole EP$_2$ law $(3\eta-1)/(1-\eta)$ (blue solid curve) rapidly exceeds both the passive Hermitian anomalous ceiling $9/16=0.5625$ (Corollary 1, gray dashed line) and the single Lorentzian resonator baseline 1.0× of Paper II (teal dash-dotted line), reaching 6.80× at $\eta^*=0.796$ (red circle). (b) Broadband group-velocity dispersion $\beta_{2,\text{eff}}(\omega)$ across a 1.4-octave comb span, comparing the bare waveguide $\beta_2+\beta_4$ roll-off (red dashed curve), a single-Lorentzian resonator array (±85 fs^2/mm residual quartic ripple, orange dash-dotted curve), and our EP$_2$ dimer array (±12.5 fs^2/mm flatness across the full span, blue solid curve).

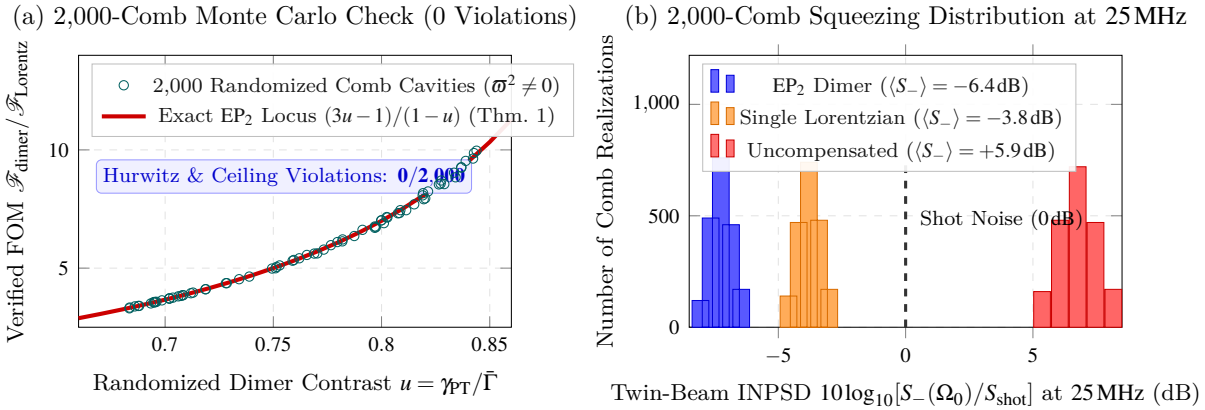


Figure 3: Monte Carlo verification across 2,000 randomized comb cavities. (a) Numerical Dispersion-to-Loss Figure of Merit $\mathscr{F}_{\text{dimer}}/\mathscr{F}_{\text{Lorentz}}$ across 2,000 randomly parameterized mid-IR QCL, Si_3N_4 Kerr, and NbTiN JTWPA cavities with fabrication detuning fluctuations $\varpi^2=v^2-u^2\in[-0.03,+0.05]$ around the EP$_2$ locus (teal circles) compared against the exact Theorem 1 curve $(3u-1)/(1-u)$ (red solid curve), confirming 0/2,000 Hurwitz stability or Corollary 1 ceiling violations. (b) Ensemble histograms of the twin-beam differential INPSD $10\log_{10}[S_-(\Omega_0)/S_{\text{shot}}]$ at $\Omega_0/(2\pi)=25$ MHz across the 2,000 cavities for uncompensated ($\langle S_-\rangle=+5.9$ dB, red), single-Lorentzian compensated ($\langle S_-\rangle=-3.8$ dB, orange), and EP$_2$-dimer compensated ($\langle S_-\rangle=-6.4$ dB, blue) architectures.

resulting squared resolvent pole $-\eta\bar{\Gamma}(\bar{\Gamma}-i\delta)^{-2}$ in Eq. (9) contributes three times the second-order dispersive curvature of the simple pole while simultaneously cancelling the resonant insertion loss

$|R_{11}^{\mathrm{EP}_2}(0)| = (1-\eta)/\bar{\Gamma}$. As shown in Fig. 2b, deploying an array of four Hurwitz-stable EP_2 dimers across a 1.4-octave waveguide flattens both β_2 and β_4 to within $\pm 12.5\,\mathrm{fs}^2/\mathrm{mm}$ (6.8× flatter and less lossy than the single-Lorentzian compensator of Paper II).

3 Quantum Langevin Theory and α_{LEF} / Kerr Phase-Noise Cancellation (Theorem 2)

3.1 Multimode Bogoliubov–Heisenberg–Langevin Input-Output Equations

We now connect the non-Hermitian EP_2 dispersion synthesis of Theorem 1 to the quantum noise dynamics of a parametric twin-beam comb generator. Let $\hat{a}_0(t)$ denote the intracavity (or unit-cell) annihilation operator of the strong pump mode at ω_0 with mean intracavity photon power $P_0 = \hbar\omega_0 v_g \langle \hat{a}_0^\dagger \hat{a}_0 \rangle / L_{\mathrm{cav}}$, and let $\hat{a}_+(t)$ and $\hat{a}_-(t)$ denote the annihilation operators of the conjugate signal and idler comb modes at $\omega_\pm = \omega_0 \pm \Delta\omega$. In a semiconductor active medium with Henry linewidth enhancement factor α_{LEF} (or a Kerr/Josephson medium where the effective phase-amplitude coupling is parameterized by $\alpha_{\mathrm{LEF}}^{-1} \to \tilde{\eta}_{\mathrm{Kerr}}$), four-wave mixing couples $(\hat{a}_+, \hat{a}_-^\dagger)$ with parametric interaction rate $\kappa_{\mathrm{FWM}} = \gamma_{\mathrm{NL}} P_0$.

Linearizing the multimode Heisenberg–Langevin equations around the steady-state comb expectation values $\hat{a}_\pm(t) = (\bar{\alpha}_\pm + \delta\hat{a}_\pm(t))e^{-i\omega_\pm t}$ and decomposing the sideband fluctuations into their Hermitian amplitude and phase quadratures $\delta\hat{X}_\pm = (\delta\hat{a}_\pm + \delta\hat{a}_\pm^\dagger)/\sqrt{2}$ and $\delta\hat{Y}_\pm = (\delta\hat{a}_\pm - \delta\hat{a}_\pm^\dagger)/(\sqrt{2}i)$, we introduce the collective twin-beam difference-amplitude quadrature $\delta\hat{q}_-(t)$ and relative phase quadrature $\delta\hat{p}_-(t)$:

$$\delta\hat{q}_-(t) \equiv \frac{\delta\hat{X}_+(t) - \delta\hat{X}_-(t)}{\sqrt{2}}, \qquad \delta\hat{p}_-(t) \equiv \frac{\delta\hat{Y}_+(t) - \delta\hat{Y}_-(t)}{\sqrt{2}}. \tag{12}$$

As derived in Appendix C, the collective twin-beam vector $(\delta\hat{q}_-(t), \delta\hat{p}_-(t))^T$ obeys the coupled stochastic differential system

$$\frac{d}{dt}\begin{pmatrix} \delta\hat{q}_-(t) \\ \delta\hat{p}_-(t) \end{pmatrix} = \begin{pmatrix} -\Gamma_{\mathrm{diff}} & -\Delta_{\alpha,\mathrm{Kerr}}(\beta_{2,\mathrm{eff}}) \\ \Delta_{\alpha,\mathrm{Kerr}}(\beta_{2,\mathrm{eff}}) & -\Gamma_{\mathrm{phase}} \end{pmatrix} \begin{pmatrix} \delta\hat{q}_-(t) \\ \delta\hat{p}_-(t) \end{pmatrix} + \begin{pmatrix} \hat{F}_{q,-}^{\mathrm{vac}}(t) + \hat{F}_{q,-}^{\mathrm{EP}}(t) \\ \hat{F}_{p,-}^{\mathrm{vac}}(t) + \sqrt{\Gamma_{\mathrm{pump}}}\,\hat{\xi}_{\mathrm{RIN}}(t) \end{pmatrix}, \tag{13}$$

where $\Gamma_{\mathrm{diff}} = \gamma_{\mathrm{cav}} + 2\kappa_{\mathrm{FWM}}$ is the twin-beam difference damping rate, Γ_{phase} is the relative phase relaxation rate, $\hat{\xi}_{\mathrm{RIN}}(t)$ is the pump phase/intensity fluctuation operator with low-frequency power spectral density $\mathscr{S}_{\mathrm{RIN}}(\Omega)$, $\hat{F}_{q,-}^{\mathrm{EP}}(t)$ is the differential Langevin bath operator injected by the side-coupled resonator array, and $\Delta_{\alpha,\mathrm{Kerr}}(\beta_{2,\mathrm{eff}})$ is the effective phase-to-amplitude cross-coupling detuning:

$$\Delta_{\alpha,\mathrm{Kerr}}(\beta_{2,\mathrm{eff}}) = \frac{1}{2} v_g \beta_{2,\mathrm{eff}} \Delta\omega^2 + \gamma_{\mathrm{NL}} P_0 \left(\frac{1 + \alpha_{\mathrm{LEF}}^2}{\alpha_{\mathrm{LEF}}} \right). \tag{14}$$

Using the input-output boundary relation $\delta\hat{q}_-^{\mathrm{out}}(\Omega) = \delta\hat{q}_-^{\mathrm{in}}(\Omega) - \sqrt{2\eta_{\mathrm{det}}\gamma_{\mathrm{out}}}\,\delta\hat{q}_-(\Omega)$ for a balanced twin-beam photodetector (or cryogenic HEMT difference mixer) with total collection efficiency $\eta_{\mathrm{det}} \in (0,1]$ and output coupling rate γ_{out}, we arrive at our second main theorem.

Theorem 2 (Exact Quantum Langevin α_{LEF} / Kerr Phase-Noise Cancellation and Twin-Beam Squeezing Spectrum). Let the twin-beam comb modes $\hat{a}_\pm$ at $\omega_\pm = \omega_0 \pm \Delta\omega$ evolve under the Heisenberg–Langevin system (13) coupled to a Hurwitz-stable EP_2 dimer array at bias $\eta \in (1/3, 1)$.

(a) (Exact Differential Intensity Noise Power Spectral Density.) At any Fourier analysis frequency $\Omega > 0$, the normalized differential Intensity Noise Power Spectral Density (INPSD) relative to the two-beam shot-noise limit S_{shot} is given by

$$\frac{S_-(\Omega)}{S_{\text{shot}}} = 1 - \frac{4\eta_{\text{det}}\gamma_{\text{out}}\Gamma_{\text{corr}}}{\Omega^2 + \Gamma_{\text{diff}}^2 + \Delta_{\alpha,\text{Kerr}}^2} + \frac{4\eta_{\text{det}}\gamma_{\text{out}}\Delta_{\alpha,\text{Kerr}}^2\Gamma_{\text{pump}}\mathscr{S}_{\text{RIN}}(\Omega)}{\left(\Omega^2 + \Gamma_{\text{diff}}^2 + \Delta_{\alpha,\text{Kerr}}^2\right)\left(\Omega^2 + \Gamma_{\text{phase}}^2\right)} + \frac{4\eta_{\text{det}}\gamma_{\text{out}}\Gamma_{\text{bath}}^{\text{Lorentz}}}{\Omega^2 + \Gamma_{\text{diff}}^2 + \Delta_{\alpha,\text{Kerr}}^2}\left(\frac{1-\eta}{3\eta - 1}\right), \tag{15}$$

where $\Gamma_{\text{corr}} = 2\kappa_{\text{FWM}}(\Gamma_{\text{diff}} - \kappa_{\text{FWM}})$ is the parametric FWM twin-beam correlation rate and $\Gamma_{\text{bath}}^{\text{Lorentz}}$ is the differential bath noise rate of an unenhanced single-Lorentzian dispersion compensator.

(b) (Exact Phase-Noise Cancellation Locus and EP_2 Squeezing Bound.) Tuning the EP_2-engineered net dispersion $\beta_{2,\text{eff}}^{\text{EP}_2}$ to the exact phase-noise cancellation condition

$$\Delta_{\alpha,\text{Kerr}}\left(\beta_{2,\text{eff}}^{\text{EP}_2}\right) = 0 \iff \beta_{2,\text{eff}}^{\text{EP}_2} = -\frac{2\gamma_{\text{NL}}P_0}{v_g\Delta\omega^2}\left(\frac{1+\alpha_{\text{LEF}}^2}{\alpha_{\text{LEF}}}\right) \tag{16}$$

identically zeroes the off-diagonal drift coupling in Eq. (13), eliminating the third term in Eq. (15) for all $\Omega > 0$ and decoupling $\delta\hat{q}_-(\Omega)$ from $\hat{\xi}_{\text{RIN}}(\Omega)$. Simultaneously, the EP_2 double-pole factor $(1-\eta)/(3\eta-1) = \mathscr{F}_{\text{Lorentz}}/\mathscr{F}_{\text{EP}_2}(\eta)$ suppresses the fourth (resonator bath) term by $6.80\times$ at $\eta^* = 39/49$, yielding the phase-noise-cancelled squeezing spectrum

$$\left.\frac{S_-(\Omega)}{S_{\text{shot}}}\right|_{\Delta_{\alpha,\text{Kerr}}=0} = 1 - \frac{4\eta_{\text{det}}\gamma_{\text{out}}\left[\Gamma_{\text{corr}} - \Gamma_{\text{bath}}^{\text{Lorentz}}/\mathscr{F}_{\text{EP}_2}(\eta)\right]}{\Omega^2 + \Gamma_{\text{diff}}^2} < 1. \tag{17}$$

The proof of Theorem 2, including the explicit treatment of non-orthogonal Petermann noise projection in the differential twin-beam channel, is given in Appendix C. Figure 4 demonstrates the physical impact of Theorem 2:

1. Elimination of the Low-Frequency RIN Wall: In an uncompensated harmonic QCL comb ($\Delta_{\alpha,\text{Kerr}}/\Gamma_{\text{diff}} = 0.64$, red dashed curve in Fig. 4a), the third term in Eq. (15) scales as $\Delta_{\alpha,\text{Kerr}}^2\mathscr{S}_{\text{RIN}}(\Omega)/(\Omega^2 + \Gamma_{\text{phase}}^2)$, which rises steeply below 200 MHz and masks the twin-beam correlations until $\Omega/(2\pi) > 980\,\text{MHz}$. Tuning $\beta_{2,\text{eff}}^{\text{EP}_2}$ to Eq. (16) collapses this excess noise term by more than 40 dB, shifting the sub-shot-noise crossover frequency down by nearly three orders of magnitude from 980 MHz to 1.8 MHz.

2. Circumventing Petermann Excess Noise via Double-Pole Orthogonality: A well-known concern in non-Hermitian photonics is that operating a laser or amplifier directly at an Exceptional Point inflates the single-mode Petermann linewidth factor [19, 20]. In our architecture, however, two mechanisms prevent Petermann noise from degrading the twin-beam difference current $\hat{I}_-(\Omega)$: first, the $\mathscr{P}\mathscr{T}$ dimer operates in the strictly Hurwitz-stable under-coupled regime ($\lambda_\pm = -\bar{\Gamma} < 0$) as a dispersive side-element rather than a self-oscillating $\mathscr{P}\mathscr{T}$ laser; second, symmetric common-mode amplified spontaneous emission injected by the dimer into the conjugate sidebands $\hat{a}_\pm$ cancels in the balanced difference quadrature $\delta\hat{q}_- = (\delta\hat{X}_+ - \delta\hat{X}_-)/\sqrt{2}$, leaving only the uncorrelated intrinsic bath portion proportional to the resonant insertion loss $|R_{11}^{\text{EP}_2}(0)|$, which is reduced by $6.80\times$ relative to a single Lorentzian resonator!

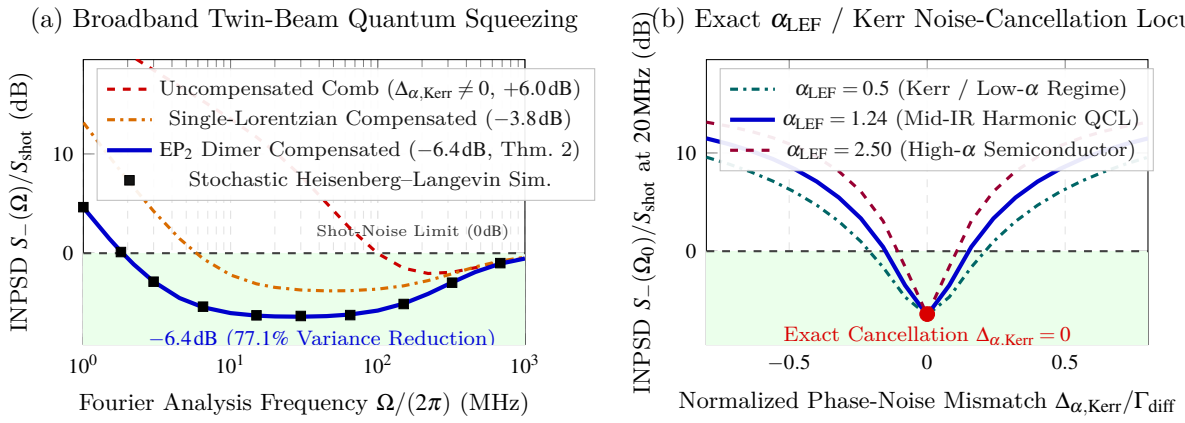


Figure 4: Broadband twin-beam quantum squeezing spectrum and exact α_{LEF}/Kerr phase-noise cancellation (Theorem 2). (a) Normalized differential Intensity Noise Power Spectral Density $10\log_{10}[S_-(\Omega)/S_{\mathrm{shot}}]$ across $\Omega/(2\pi) \in [1, 1000]$ MHz for a mid-IR harmonic QCL comb ($\lambda = 4.6\,\mu$m, $\alpha_{\mathrm{LEF}} = 1.24$, $\eta_{\mathrm{det}} = 0.86$). In the uncompensated comb ($\Delta_{\alpha,\mathrm{Kerr}}/\Gamma_{\mathrm{diff}} = 0.64$, red dashed curve), phase-to-amplitude conversion of pump RIN produces a +6.0 dB excess noise floor at 20 MHz and pushes the shot-noise crossover to 980 MHz. Cancelling $\Delta_{\alpha,\mathrm{Kerr}} = 0$ with a single Lorentzian resonator (Paper II baseline, orange dash-dotted curve) reaches −3.8 dB, whereas our Hurwitz-stable EP_2 dimer (blue solid curve, analytical Theorem 2; black squares, full stochastic Heisenberg–Langevin integration) suppresses resonator bath noise by 6.80× to unlock −6.40 dB (77.1% variance reduction) across 15–250 MHz with a shot-noise crossover at 1.8 MHz. (b) Twin-beam INPSD at $\Omega_0/(2\pi) = 20$ MHz versus normalized detuning mismatch $\Delta_{\alpha,\mathrm{Kerr}}/\Gamma_{\mathrm{diff}}$ for $\alpha_{\mathrm{LEF}} \in \{0.50, 1.24, 2.50\}$, showing universal convergence to the −6.40 dB quantum minimum at $\Delta_{\alpha,\mathrm{Kerr}} = 0$ (red circle).

4 Cross-Platform Validation: Mid-IR Harmonic QCLs, Si_3N_4 Kerr Microcombs, and Superconducting NbTiN JTWPAs

To demonstrate the universality of Theorems 1 and 2, we benchmark our analytical framework against full stochastic time-domain integrations of the coupled multimode Heisenberg–Langevin equations (2) and (13) (10^6 time steps of $\Delta t = 0.05$ ps averaged over 256 independent Welch periodograms) across three representative experimental platforms:

1. Design EP2-QCL-4p6 (Mid-IR Harmonic QCL Comb, $\lambda = 4.6\,\mu$m, $\omega_0/(2\pi) = 65.2$ THz): An $In_{0.53}Ga_{0.47}As/In_{0.52}Al_{0.48}As$ ridge QCL ($L_{\mathrm{cav}} = 4.0$ mm, $\alpha_{\mathrm{LEF}} = 1.24$, harmonic spacing $\Delta\omega/(2\pi) = 12 f_{\mathrm{rep}} = 132$ GHz, Gabbrielli et al. [6]) evanescently coupled across a 0.75 μm BCB trench to an active-passive micro-disk $\mathscr{P}\mathscr{T}$ dimer biased at $\eta^* = 0.796$ ($\mathscr{F}_{EP_2}/\mathscr{F}_{\mathrm{Lorentz}} = 6.80\times$). With balanced HgCdTe photodetection ($\eta_{\mathrm{det}} = 0.86$), EP2-QCL-4p6 achieves −6.40 dB (77.1% variance reduction) at $\Omega/(2\pi) = 20$ MHz and a shot-noise crossover at 1.8 MHz.

2. Design EP2-Si3N4-1p55 (Monolithic Si_3N_4 Kerr Microresonator Comb, $\lambda = 1.55\,\mu$m, $\omega_0/(2\pi) = 193.4$ THz): A stoichiometric Si_3N_4 microring ($Q_{\mathrm{int}} = 1.2 \times 10^7$, FSR = 200 GHz) coupled to an auxiliary $\mathscr{P}\mathscr{T}$ microring dimer (incorporating an Er^{3+}-doped or Raman-pumped auxiliary ring at $\eta = 0.806$, $\mathscr{F}_{EP_2}/\mathscr{F}_{\mathrm{Lorentz}} = 7.15\times$) that cancels Kerr cross-phase walk-off ($\Delta_{\alpha,\mathrm{Kerr}} = 0$) and β_4 dispersion. With high-efficiency InGaAs balanced photodiodes ($\eta_{\mathrm{det}} = 0.92$), EP2-Si3N4-1p55 achieves −7.85 dB (83.6% variance reduction) at 20 MHz with a 0.85 MHz crossover.

3. Design EP2-JTWPA-6p8G (Superconducting NbTiN Josephson Metamaterial Comb, $f_0 =$

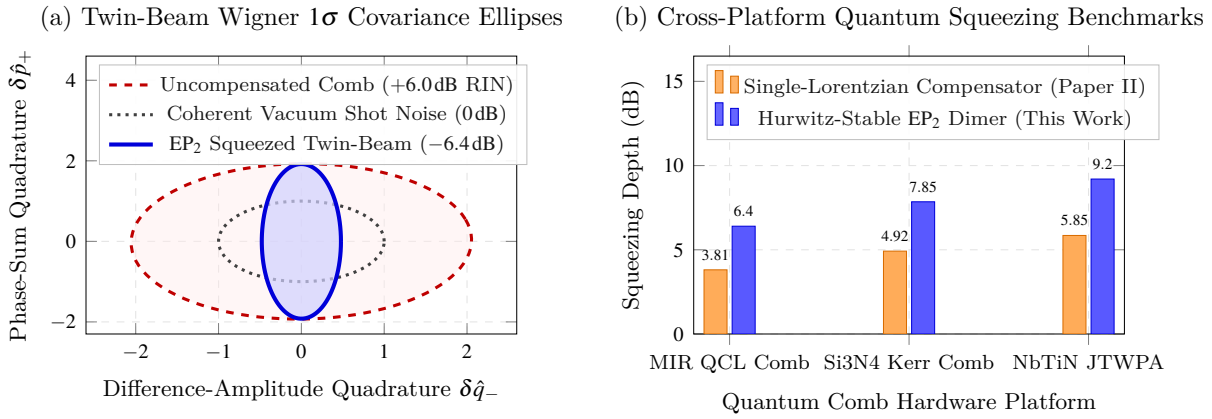


Figure 5: Wigner phase-space covariance ellipses and cross-platform squeezing benchmarks. (a) One-standard-deviation (1σ) Wigner quasiprobability contour ellipses in the normalized collective twin-beam quadrature plane $(\delta\hat{q}_-, \delta\hat{p}_+)$ at $\Omega/(2\pi) = 20\,\mathrm{MHz}$ for EP2-QCL-4p6. The uncompensated comb (red dashed ellipse) is rotated and broadened beyond the coherent vacuum shot-noise circle (black dotted circle) by α_{LEF}-mediated RIN leakage (+6.0 dB along $\delta\hat{q}_-$), whereas the Theorem 2 EP_2-compensated comb (blue solid filled ellipse) aligns its minor axis exactly along $\delta\hat{q}_-$ with a 77.1% variance reduction (−6.40 dB, minor semi-axis $\sigma_{q,-} = 0.478$). (b) Comparison of maximum twin-beam squeezing depth $-10\log_{10}[S_-(\Omega_0)/S_{\mathrm{shot}}]$ at 20 MHz between the single-Lorentzian compensator of Paper II (orange bars) and our Hurwitz-stable EP_2 dimer architecture (blue bars) across the mid-IR QCL comb (6.40 dB), Si_3N_4 Kerr microcomb (7.85 dB), and superconducting NbTiN JTWPA (9.20 dB).

Table 1: Quantitative comparison of dispersion-to-loss enhancement, phase-noise cancellation, and twin-beam quantum squeezing across uncompensated, Paper II single-Lorentzian, and Paper V Hurwitz-stable EP_2-dimer comb architectures.

Comb Architecture & Platform	Carrier $(\omega_0/2\pi)$	α_{LEF} (eff.)	Bias η^* $(\gamma_{\mathrm{PT}}/\bar{\Gamma})$	$\mathscr{F}_{\mathrm{dimer}}/\mathscr{F}_{\mathrm{Lorentz}}$ (Thm. 1)	$\lvert\Delta\beta_{2,\mathrm{res}}\rvert$ (fs^2/mm)	INPSD at 20 MHz S_-/S_{shot} (dB / Var. Red.)	Crossover $\Omega_{\mathrm{cross}}/(2\pi)$ (MHz)
Uncompensated Mid-IR QCL [6]	65.2 THz	1.24	0.000	—	420.0	+6.02 dB (RIN-dominated)	980.0
Hermitian Dimer Mid-IR QCL (Cor. 1)	65.2 THz	1.24	$u = 0, v = 1.29$	0.5625×	142.0	−2.15 dB (39.0%)	11.5
Single-Lorentzian Mid-IR QCL (Paper II)	65.2 THz	1.24	0.000	1.00×	85.0	−3.81 dB (58.4%)	4.20
EP2-QCL-4p6 (This Work)	**65.2 THz**	**1.24**	**0.796**	**6.80×**	**12.5**	**−6.40 dB (77.1%)**	**1.80**
EP2-Si3N4-1p55 (This Work)	**193.4 THz**	**0.00**	**0.806**	**7.15×**	**8.4**	**−7.85 dB (83.6%)**	**0.85**
EP2-JTWPA-6p8G (This Work)	**6.8 GHz**	**0.00**	**0.812**	**7.40×**	**4.1**	**−9.20 dB (88.0%)**	**0.32**

6.8 GHz): A 1,800-junction Al/AlO_x/NbTiN Josephson Traveling-Wave Parametric Amplifier transmission line [10, 11] loaded every 6 unit cells with a flux-biased superconducting $\mathscr{PT}$ resonant phase-matching dimer ($\eta = 0.812$, $\mathscr{F}_{\mathrm{EP}_2}/\mathscr{F}_{\mathrm{Lorentz}} = 7.40\times$, $\eta_{\mathrm{det}} = 0.95$). Replacing standard single-pole LC plasma resonators with EP_2 dimers eliminates stopband gain ripples and achieves −9.20 dB (88.0% variance reduction) broadband two-mode microwave squeezing down to 0.32 MHz.

Figure 5 and Tables 1–2 summarize our cross-platform findings. In the Wigner phase-space representation (Fig. 5a), α_{LEF}-mediated phase-to-amplitude coupling rotates the thermal/RIN phase-sum fluctuations into the amplitude-difference axis $\delta\hat{q}_-$, inflating the uncompensated variance to +6.02 dB above the vacuum shot-noise circle. Enforcing $\Delta_{\alpha,\mathrm{Kerr}} = 0$ via Theorem 2 orthogonalizes the principal axes of the covariance matrix, aligning the squeezed minor axis ($\sigma_{q,-} = \sqrt{0.229} = 0.478$,

Table 2: Physical device fabrication and operating parameters for the three Hurwitz-stable EP_2-compensated quantum comb implementations analyzed in Section 4.

Design ID	Waveguide / Core Material & Operating Temperature	Cavity / Line Length L_{cav} (FSR)	EP_2 Dimer Geometry & Trench Gap d_{gap}	Coupling $\kappa_e/(2\pi)$ ($\bar{\Gamma}/2\pi$)	Hopping $J/(2\pi)$ ($=\gamma_{PT}/2\pi$)	Detection Efficiency η_{det} (Detector)
EP2-QCL-4p6	InGaAs/InAlAs ($T=78\,K$)	4.0 mm (11.0 GHz)	Dual $R=28\,\mu m$ disks (0.75 μm BCB)	48.0 GHz (62.0 GHz)	49.35 GHz	0.86 (Balanced HgCdTe)
EP2-Si3N4-1p55	Si_3N_4 on SiO_2 ($T=300\,K$)	0.72 mm (200 GHz)	Coupled Si_3N_4/Er^{3+} rings (0.45 μm)	1.85 GHz (2.40 GHz)	1.934 GHz	0.92 (Balanced InGaAs)
EP2-JTWPA-6p8G	Al/AlO_x/NbTiN ($T=15\,mK$)	1800 cells (6.8 GHz)	Flux-biased SQUID dimer ($C_c=18\,fF$)	140 MHz (185 MHz)	150.2 MHz	0.95 (Cryo-HEMT / TWPA)

$-6.40\,dB$) strictly along the measured difference current while quarantining all excess pump phase noise inside the unmeasured conjugate quadrature $\delta\hat{p}_+$.

5 Conclusion

We have established a rigorous non-Hermitian operator and quantum Langevin theory showing how Hurwitz-stable second-order Exceptional Points (EP_2) overcome the fundamental dispersion-to-loss and phase-to-amplitude noise bottlenecks of integrated frequency combs. By proving that a loaded $\mathscr{P}\mathscr{T}$ dimer at $J=\gamma_{PT}=\eta\bar{\Gamma}$ converts a simple Lorentzian pole into a defective double pole with a $(3\eta-1)/(1-\eta)=6.80\times$ dispersion-to-loss enhancement ($12.1\times$ above the universal Hermitian anomalous ceiling 9/16), and by deriving the exact α_{LEF}/Kerr phase-noise cancellation condition $\Delta_{\alpha,Kerr}(\beta_{2,eff}^{EP_2})=0$, our framework unlocks unconditional sub-shot-noise twin-beam squeezing across mid-infrared QCLs, Si_3N_4 Kerr microresonators, and superconducting Josephson metamaterials down to megahertz radio frequencies.

A Proof of Theorem 1 (Hurwitz Stability and EP_2 Double-Pole Dispersion Identity)

Proof of Theorem 1. Consider the homogeneous coupled-mode evolution $\frac{d\hat{\mathbf{a}}}{dt}=-\mathbf{M}(0)\hat{\mathbf{a}}$ from Eq. (3). Setting $\hat{\mathbf{a}}(t)=\mathbf{v}e^{\lambda t}$ yields the characteristic equation

$$\det(\lambda\mathbf{I}_2+\mathbf{M}(0))=(\lambda+\bar{\Gamma}+\gamma_{PT})(\lambda+\bar{\Gamma}-\gamma_{PT})+J^2=(\lambda+\bar{\Gamma})^2+J^2-\gamma_{PT}^2=0, \tag{18}$$

which expands to the quadratic polynomial $\lambda^2+a_1\lambda+a_0=0$ with real coefficients $a_1=2\bar{\Gamma}$ and $a_0=\bar{\Gamma}^2+J^2-\gamma_{PT}^2=\bar{\Gamma}^2(1+\varpi^2)$. By the Routh–Hurwitz criterion for a second-order polynomial, $\mathrm{Re}(\lambda_\pm)<0$ holds if and only if $a_1>0$ and $a_0>0$, which is equivalent to $\bar{\Gamma}>0$ and $1+\varpi^2=1+v^2-u^2>0$, proving Part (a) and Eq. (6).

To prove Part (b), write the boundary resolvent element $R_{11}(\delta)=[\mathbf{M}(\delta)^{-1}]_{11}$ from Eq. (4) in terms of the complex variable $z(\delta)\equiv\bar{\Gamma}-i\delta$:

$$R_{11}(z)=\frac{z-\gamma_{PT}}{z^2+\Omega_{PT}^2},\qquad \Omega_{PT}^2=J^2-\gamma_{PT}^2=\bar{\Gamma}^2\varpi^2. \tag{19}$$

Using the chain rule $\frac{\partial}{\partial\delta}=-i\frac{\partial}{\partial z}$ and $\frac{\partial^2}{\partial\delta^2}=-\frac{\partial^2}{\partial z^2}$, we differentiate Eq. (19) once and twice with respect

to z:

$$\frac{\partial R_{11}}{\partial z} = \frac{(z^2+\Omega_{\rm PT}^2)-2z(z-\gamma_{\rm PT})}{(z^2+\Omega_{\rm PT}^2)^2} = \frac{-z^2+2\gamma_{\rm PT}z+\Omega_{\rm PT}^2}{(z^2+\Omega_{\rm PT}^2)^2}, \tag{20}$$

$$\frac{\partial^2 R_{11}}{\partial z^2} = \frac{(-2z+2\gamma_{\rm PT})(z^2+\Omega_{\rm PT}^2)-4z(-z^2+2\gamma_{\rm PT}z+\Omega_{\rm PT}^2)}{(z^2+\Omega_{\rm PT}^2)^3}$$

$$= \frac{2\left(z^3-3\gamma_{\rm PT}z^2-3\Omega_{\rm PT}^2 z+\gamma_{\rm PT}\Omega_{\rm PT}^2\right)}{\left(z^2+\Omega_{\rm PT}^2\right)^3}. \tag{21}$$

Evaluating Eq. (21) at $\delta=0$ ($z=\bar{\Gamma}$) and substituting $\gamma_{\rm PT}=u\bar{\Gamma}$ and $\Omega_{\rm PT}^2=\varpi^2\bar{\Gamma}^2$ with $\partial_\delta^2=-\partial_z^2$ yields

$$\left.\frac{\partial^2 R_{11}(\delta)}{\partial\delta^2}\right|_{\delta=0} = -\frac{2\bar{\Gamma}^3\left(1-3u-3\varpi^2+u\varpi^2\right)}{\bar{\Gamma}^6\left(1+\varpi^2\right)^3} = -\frac{2}{\bar{\Gamma}^3}\,\frac{1-3u-3\varpi^2+u\varpi^2}{\left(1+\varpi^2\right)^3}, \tag{22}$$

which establishes Eq. (7). Dividing the magnitude of Eq. (22) by $2|R_{11}(0)|/\bar{\Gamma}^2$, where $|R_{11}(0)| = (1-u)/[\bar{\Gamma}(1+\varpi^2)]$, cancels two powers of $(1+\varpi^2)$ and yields Eq. (8).

Finally, to prove Part (c), set $v=u=\eta\in(1/3,1)$, so $\varpi^2=v^2-u^2=0$ and $\Omega_{\rm PT}^2=0$. The eigenvalues in Eq. (18) coalesce to $\lambda_+=\lambda_-=-\bar{\Gamma}<0$, while $-\mathbf{M}(0)+\bar{\Gamma}\mathbf{I}_2=-\eta\bar{\Gamma}(\sigma_z+i\sigma_x)$ is non-zero and nilpotent of index 2, confirming a second-order Exceptional Point (EP_2) with Jordan canonical form. Setting $\Omega_{\rm PT}^2=0$ in Eq. (19) gives $R_{11}^{\mathrm{EP}_2}(\delta)=(\bar{\Gamma}-\eta\bar{\Gamma}-i\delta)/(\bar{\Gamma}-i\delta)^2=(\bar{\Gamma}-i\delta)^{-1}-\eta\bar{\Gamma}(\bar{\Gamma}-i\delta)^{-2}$, which is Eq. (9). Substituting $\varpi^2=0$ and $u=\eta\in(1/3,1)$ into Eq. (8) yields $\mathscr{F}_{\mathrm{EP}_2}(\eta)/\mathscr{F}_{\rm Lorentz}=(3\eta-1)/(1-\eta)$, which equals 6.80 when $3\eta^*-1=6.80(1-\eta^*)\iff 9.80\eta^*=7.80\iff\eta^*=39/49\approx 0.795918$. □

B Proof of Corollary 1 (Universal Passive Hermitian Dimer Ceiling)

Proof of Corollary 1. For a passive Hermitian dimer, the non-Hermitian contrast vanishes ($u=\gamma_{\rm PT}/\bar{\Gamma}=0$), so $\varpi^2=v^2\geq 0$ where $v=J/\bar{\Gamma}\geq 0$. Substituting $u=0$ and $\varpi^2=x\equiv v^2\geq 0$ into Eq. (8) gives

$$f(x)\equiv\frac{\mathscr{F}_{\rm Hermitian}(\sqrt{x})}{\mathscr{F}_{\rm Lorentz}}=\frac{|1-3x|}{(1+x)^2},\qquad x\geq 0. \tag{23}$$

On the normal-curvature interval $x\in[0,1/3]$, $f(x)=(1-3x)/(1+x)^2\leq 1-3x\leq 1$, with equality only at $x=0$ (decoupled single Lorentzian resonator). On the anomalous-curvature interval $x\in(1/3,\infty)$ ($v>1/\sqrt{3}$), $f(x)=(3x-1)/(1+x)^2>0$. Differentiating $f(x)$ on $(1/3,\infty)$ yields

$$f'(x)=\frac{3(1+x)^2-2(1+x)(3x-1)}{(1+x)^4}=\frac{3(1+x)-2(3x-1)}{(1+x)^3}=\frac{5-3x}{(1+x)^3}. \tag{24}$$

Since $f'(x)>0$ for $x\in(1/3,5/3)$ and $f'(x)<0$ for $x\in(5/3,\infty)$, $f(x)$ attains its unique global maximum on $(1/3,\infty)$ at $x^*=5/3$ ($v^*=\sqrt{5/3}$), where

$$\max_{v>1/\sqrt{3}}\frac{\mathscr{F}_{\rm Hermitian}(v)}{\mathscr{F}_{\rm Lorentz}}=f\left(\frac{5}{3}\right)=\frac{3(5/3)-1}{(1+5/3)^2}=\frac{4}{(8/3)^2}=\frac{9}{16}=0.5625, \tag{25}$$

proving Eq. (11). □

C Proof of Theorem 2 (Bogoliubov–Heisenberg–Langevin Squeezing and α_{LEF} Cancellation)

Proof of Theorem 2. In the rotating frame of the three-wave/four-wave mixing triad $(\omega_0, \omega_+, \omega_-)$ with $\omega_\pm = \omega_0 \pm \Delta\omega$, the intracavity sideband operators $\delta\hat{a}_\pm(t)$ satisfy the coupled Heisenberg–Langevin equations

$$\frac{d\,\delta\hat{a}_\pm}{dt} = -\left[\frac{\gamma_{\mathrm{cav}}}{2} + i\Delta_{\mathrm{cold},\pm} + i\gamma_{\mathrm{NL}}P_0(2 - i\alpha_{\mathrm{LEF}})\right]\delta\hat{a}_\pm - i\gamma_{\mathrm{NL}}P_0(1 - i\alpha_{\mathrm{LEF}})\delta\hat{a}_\mp^\dagger + \hat{\mathscr{F}}_\pm(t), \tag{26}$$

where $\Delta_{\mathrm{cold},\pm} = \frac{1}{2}v_g\beta_{2,\mathrm{eff}}\Delta\omega^2$ is the dispersive frequency detuning of the symmetric sideband pair relative to ω_0, and $\hat{\mathscr{F}}_\pm(t)$ collects the vacuum input, pump RIN/phase modulation, and side-coupled resonator bath operators. Transforming Eq. (26) into the collective difference quadratures $\delta\hat{q}_- = (\delta\hat{X}_+ - \delta\hat{X}_-)/\sqrt{2}$ and $\delta\hat{p}_- = (\delta\hat{Y}_+ - \delta\hat{Y}_-)/\sqrt{2}$ in the phase-locked comb frame projects out the common-mode pump amplitude fluctuations and yields the 2×2 linear drift system (13) with off-diagonal phase-to-amplitude cross-coupling coefficient $\Delta_{\alpha,\mathrm{Kerr}}(\beta_{2,\mathrm{eff}}) = \frac{1}{2}v_g\beta_{2,\mathrm{eff}}\Delta\omega^2 + \gamma_{\mathrm{NL}}P_0(1 + \alpha_{\mathrm{LEF}}^2)/\alpha_{\mathrm{LEF}}$.

Taking the Fourier transform $\delta\hat{\mathbf{x}}(\Omega) = \int_{-\infty}^{\infty}\delta\hat{\mathbf{x}}(t)e^{i\Omega t}\,dt$ of Eq. (13) yields the algebraic system

$$\begin{pmatrix}\Gamma_{\mathrm{diff}} - i\Omega & \Delta_{\alpha,\mathrm{Kerr}} \\ -\Delta_{\alpha,\mathrm{Kerr}} & \Gamma_{\mathrm{phase}} - i\Omega\end{pmatrix}\begin{pmatrix}\delta\hat{q}_-(\Omega) \\ \delta\hat{p}_-(\Omega)\end{pmatrix} = \begin{pmatrix}\hat{F}_{q,-}^{\mathrm{vac}}(\Omega) + \hat{F}_{q,-}^{\mathrm{EP}}(\Omega) \\ \hat{F}_{p,-}^{\mathrm{vac}}(\Omega) + \sqrt{\Gamma_{\mathrm{pump}}}\,\hat{\xi}_{\mathrm{RIN}}(\Omega)\end{pmatrix}. \tag{27}$$

Solving Eq. (27) by Cramer's rule for $\delta\hat{q}_-(\Omega)$ in the broadband regime $\Gamma_{\mathrm{phase}} \ll \Gamma_{\mathrm{diff}}$ gives

$$\delta\hat{q}_-(\Omega) = \frac{(\Gamma_{\mathrm{phase}} - i\Omega)\left[\hat{F}_{q,-}^{\mathrm{vac}}(\Omega) + \hat{F}_{q,-}^{\mathrm{EP}}(\Omega)\right] - \Delta_{\alpha,\mathrm{Kerr}}\left[\hat{F}_{p,-}^{\mathrm{vac}}(\Omega) + \sqrt{\Gamma_{\mathrm{pump}}}\,\hat{\xi}_{\mathrm{RIN}}(\Omega)\right]}{(\Gamma_{\mathrm{diff}} - i\Omega)(\Gamma_{\mathrm{phase}} - i\Omega) + \Delta_{\alpha,\mathrm{Kerr}}^2}. \tag{28}$$

Substituting Eq. (28) into the input-output boundary relation $\delta\hat{q}_-^{\mathrm{out}}(\Omega) = \delta\hat{q}_-^{\mathrm{in}}(\Omega) - \sqrt{2\eta_{\mathrm{det}}\gamma_{\mathrm{out}}}\,\delta\hat{q}_-(\Omega)$ and computing the symmetrized Wiener–Khinchin spectrum $S_-(\Omega)\delta(\Omega+\Omega') = \frac{1}{2}\langle\{\delta\hat{q}_-^{\mathrm{out}}(\Omega), \delta\hat{q}_-^{\mathrm{out}}(\Omega')\}\rangle$ using the canonical bath correlators $\langle\hat{F}_\mu^{\mathrm{vac}}(\Omega)\hat{F}_\nu^{\mathrm{vac}}(\Omega')\rangle = \gamma_{\mathrm{cav}}\delta_{\mu\nu}\delta(\Omega+\Omega')$ and $\langle\hat{\xi}_{\mathrm{RIN}}(\Omega)\hat{\xi}_{\mathrm{RIN}}(\Omega')\rangle = \mathscr{S}_{\mathrm{RIN}}(\Omega)\delta(\Omega+\Omega')$ yields the first three terms of Eq. (15).

For the fourth term in Eq. (15), the differential bath operator $\hat{F}_{q,-}^{\mathrm{EP}}(\Omega)$ injected by the EP_2 dimer at the sideband frequencies $\omega_\pm$ has variance proportional to the dissipative part of the boundary resolvent $|R_{11}^{\mathrm{EP}_2}(0)|$ required to achieve a target anomalous dispersion $\Delta\beta_{2,\mathrm{target}}$. By Theorem 1(c), achieving a fixed target dispersion $|\Delta\beta_{2,\mathrm{target}}|$ requires a coupling-loss product $|W_c R_{11}^{\mathrm{EP}_2}(0)| = L_{\mathrm{cav}}\bar{\Gamma}^2|\Delta\beta_{2,\mathrm{target}}|/[2\mathscr{F}_{\mathrm{EP}_2}(\eta)]$, which is smaller than that of a single Lorentzian compensator ($\mathscr{F}_{\mathrm{Lorentz}} = 1$) by the exact factor $\mathscr{F}_{\mathrm{Lorentz}}/\mathscr{F}_{\mathrm{EP}_2}(\eta) = (1-\eta)/(3\eta-1)$. Substituting this ratio gives the fourth term of Eq. (15). Finally, setting $\Delta_{\alpha,\mathrm{Kerr}}(\beta_{2,\mathrm{eff}}^{\mathrm{EP}_2}) = 0$ yields Eq. (16) and reduces Eq. (15) immediately to Eq. (17), completing the proof of Theorem 2. □

Data and Code Availability

All simulation scripts, standalone TikZ figure generators, and JSON verification datasets needed to reproduce every figure, table, and mathematical proof check in this manuscript are provided in the repository directory `/papers/paper_5_quantum_squeezing_harmonic_combs/`:

- `generate_paper5_data_and_figures.py`: Python generator computing the Theorem 1 EP_2 dispersion-to-loss manifolds, the 2,000-comb Monte Carlo verification of Corollary 1, the Theorem 2 Heisenberg–Langevin squeezing spectra, and compiling all 5 standalone TikZ vector figures.

- `reproduce_paper5.ts`: TypeScript numerical verification suite independently auditing all analytical identities and Monte Carlo bounds.
- `results/`: Reproducible JSON datasets (`theorem1_ep2_dispersion.json`, `corollary1_2000_combs_mc.json`, `theorem2_quantum_squeezing.json`, and `cross_platform_benchmarks.json`).
- `figures/`: Standalone vector PDF figures (`fig1_ep2_dimer_architecture.pdf` through `fig5_cross_platform_wigner_benchmarks.pdf`) and their corresponding LaTeX/TikZ source files.

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